

MSDS 535 – Week 12

Chapter 12

Overview and Objectives

- Compare and Contrast Clustering- vs. classification-based approaches.
- Interpret Mining contextual and collective outliers.
- Detect Outlier detection in high-dimensional data.
- Summarize Outlier detection.

Data mining trends and research frontiers

- Data mining, a relatively young field since the 1980s, has seen significant progress and widespread application.
- Ubiquitous in various areas with numerous commercial systems and services available.
- Despite advancements, many challenges in data mining persist.
- The final chapter outlines future trends and research frontiers in data mining.

Chapter 12

- Section 12.1 covers methodologies for mining complex data types such as text, graphs, networks, and spatiotemporal data.
- Section 12.2 explores data mining applications, including sentiment analysis, truth discovery, misinformation identification, information and disease propagation, and productivity and team science.
- Section 12.3 introduces other data mining approaches, including structuring unstructured data, data augmentation, causality analysis, network-as-a-context, and auto-ML.
- Social impacts of data mining are discussed in Section 12.4.

Mining rich data types – Text data

- Data can be structured, semistructured, or unstructured, with most data being unstructured, including text and rich media formats.
- Text mining extracts valuable information from text, covering tasks like entity extraction, relation extraction, sentiment analysis, and document summarization.
- Text mining methods include supervised, semisupervised, distantly supervised, weakly supervised, and self-supervised approaches.

Mining rich data types – Text data

- Representation learning, crucial for text mining, focuses on distributed representation using neural networks to address data sparsity and high dimensionality.
- Word embedding methods, like Word2vec, encode words in a lower-dimensional vector space, capturing semantic meaning based on context.
- Various word embedding methods, such as GloVe and fastText, use different techniques like embeddings in hyperbolic space and user-guided topic embedding.

Mining rich data types – Text data

- Pretrained language models (PLMs), like BERT, use larger corpora and parameters, achieving high performance in natural language processing.
- PLMs use a pretraining-and-fine-tuning pipeline, advancing natural language processing and text mining.
- Text mining tasks include information extraction (NER, coreference resolution, relationship extraction), taxonomy construction, text clustering, topic modeling, and text classification.
- Text summarization involves extraction or abstraction, with recent studies proposing extract-then-rewrite methods and machine-aided human summarization.

Spatial-temporal data

- Space and time are vital in applications like smart cities, epidemiology, and astronomy.
- Data mining principles, including those in this book, can be adapted for spatial-temporal data.
- Unique properties of spatial-temporal data include auto-correlation and heterogeneity.
- Auto-correlation means dependencies in nearby locations and time, while heterogeneity refers to nonstationary distribution.

Spatial-temporal data

- Auto-correlation in spatial-temporal data poses challenges and opportunities for sophisticated modeling in data mining techniques.
- Spatial-temporal data types encompass events, trajectories, point reference data, and raster data, representing various semantics.
- Event data involves operations like intersections and similarity/difference, applied in analyzing forest fires and rainfall effects on traffic.

Spatial-temporal data

- Trajectories represent object paths over space and time, used in applications like transportation and epidemiology.
- Point reference data, collected at discrete locations, is used for reconstructing fields and modeling nonstationary spatial random processes.
- Raster data, recorded at fixed locations and time points, is employed in applications like medical imaging.

Spatial-temporal data

- Spatial and temporal data models include object, field, and spatial network models for spatial data, and temporal snapshot, temporal change, and event/process models for temporal data.
- Incorporating temporal data involves models like the temporal snapshot model, temporal change model, and event/process model.
- Rich spatial-temporal data types and models require adaptations of generic data mining methods and the development of specific methods for mining spatial-temporal relationships.

Graph and networks

- The world is more connected than ever, with complex systems modeled as graphs or networks.
- Graphs can be plain or attributed, static or dynamic, homogeneous or heterogeneous based on different aspects.
- Examples of homogeneous and heterogeneous networks include collaboration and bibliographic networks.

Graph and networks

- Graph mining explores insights and patterns through graph modeling, heterogeneous network mining, and knowledge graph mining.
- Graph modeling involves understanding graph creation mechanisms, with models like ER graph, realistic graph generators, and deep graph generative models.
- Heterogeneous network mining uses metapaths to capture rich semantics, benefiting tasks like similarity search, classification, clustering, and network embedding.

Graph and networks

- Existing methods on heterogeneous networks often require manual definition of meaningful metapaths, posing challenges in effective model design.
- Knowledge graph, a directed heterogeneous multigraph, links concepts and entities, used in applications like question answering and fact checking.
- Constructing knowledge graphs faces challenges, and recent methods explore automatic construction using language models.

Graph and networks

- Data mining tasks on knowledge graphs include link prediction, entity resolution, and triplet classification, often based on representation learning.
- Despite extensive research, challenges remain in achieving high-precision knowledge graph construction and dynamic updates.

Data mining for sentiment and opinion

- Data mining on user-produced data, like e-commerce product reviews, involves sentiment analysis and opinion mining.
- Sentiment is the view or attitude toward a situation, while opinion is a view or judgment not necessarily based on fact.
- Sentiment analysis and opinion mining are interdisciplinary, involving natural language processing, classification, clustering, and outlier analysis.
- Machine learning and data mining techniques are applied, leveraging insights from psychology and social sciences.

Data mining for sentiment and opinion

- Sentiment analysis and opinion mining is dynamic, adapting to new media types and expressions, adopting the latest data mining developments.
- Technical directions explored include sentiment/opinion analysis at different levels, lexicon-based classification, analysis of stand-alone posts, and intent mining.
- Mining intent is challenging and related to sentiment analysis, while detecting opinion spam involves techniques like network mining.

Data mining for sentiment and opinion

- Applications include understanding product advantages/disadvantages, stock market prediction, and monitoring public sentiments and opinions for government administration.
- Challenges in real-time sentiment analysis for automation systems include efficiency, accuracy, and robustness, facing noise and potential malicious attacks.
- Public sentiment monitoring in electoral politics or fighting against cyber-violence raises new challenges like fairness and privacy protection.

Truth discovery

- In data mining, truth discovery addresses conflicting values for the same data item from different sources.
- Two subclasses: single-truth (one true value allowed) and multitrueth (multiple true values allowed).
- Multitrueth considers the reliability of sources providing more correct values and less incorrect values.

Truth discovery

- Early methods used a voting system, but unreliable sources can distort results; algorithms must account for source reliability.
- Supervised learning for source reliability is unrealistic due to the vast number of sources; weakly supervised or self-supervised methods are more practical.
- Truth discovery involves computing source trustworthiness and claim reliability iteratively until convergence.

Truth discovery

- Detecting copying behaviors is crucial; systems decrease the weight of votes associated with copied values.
- Different trustworthiness for various data categories may be considered.
- For multitruth discovery, evaluation considers multiple truth values; methods include iterative mutual enhancement and probabilistic graphical models.
- Confidence in numerical value-based facts is evaluated based on data distribution; deviation from the norm indicates potential misinformation.

Identification of Misinformation

- Misinformation is false, inaccurate, or misleading information deliberately created and propagated.
- Social media is a significant platform for misinformation spread due to lack of regulation.
- Similar principles to truth discovery can be applied for misinformation identification, considering mutual enhancement between source trustworthiness and claim credibility.

Identification of Misinformation

- Consistent information from multiple trustworthy sources is considered credible, while conflicting information from less trustworthy sources is likely misinformation.
- Massive social media requires additional vigilance; simple votes are insufficient due to potential repetition and group biases.
- Methods for identifying misinformation include natural language processing, social network analysis, and algorithmic fact checkers.

Identification of Misinformation

- Efforts include developing information and media literacy, critical thinking, and avoiding biased reasoning.
- Effective software programs are deployed to detect and alert likely misinformation and provide supplemental information for fact-checking.

Information and disease propagation

- Data mining crucial in understanding information and disease propagation with unprecedented digital data scale and speed.
- Research problems: prediction (popularity, spread) and optimization (influence, source identification) based on propagation models.
- Applications include predicting virality in social media and identifying critical network conditions in computational epidemiology.

Information and disease propagation

- Prediction scenarios cover classification/regression, pre/post-publication, and different granularities (individuals, groups).
- Optimization problems include influence maximization, source localization, activity shaping, and graph connectivity.
- Models for information diffusion: feature-based, generative (Poisson, survival, Hawkes), deep learning, and computational epidemiology.

Information and disease propagation

- Computational epidemiology models (SIS, SIR, SEIR) for disease spread with challenges in complexity and estimation.
- Fast-growing research with challenges in epidemics modeling, source localization, and activity shaping.
- Questions remain about predictability and interpretability of propagation models in various scenarios.

Productivity and team science

- Increasing interest in understanding team performance, member productivity, and content impact.
- Teams, defined by roles, crucial for achieving goals in modern organizations, often embedded in networks.
- Real-world teams in film, sports, research, with attributes and networks (e.g., movie crew, coauthorship).

Productivity and team science

- Data mining techniques address key questions in team science: characterization, prediction, optimization, explanation.
- Team performance characterization involves team leader, members, environment, and task patterns.
- Team performance prediction requires identifying features, modeling correlation, and encoding team evolution dynamics.

Productivity and team science

- Team performance optimization involves optimizing team composition considering skills and network connectivity.
- Team performance explanation is challenging; sparse literature uses influence function to interpret results.
- Sparse literature on explainable team performance often uses influence functions for interpretation.

Productivity and team science

- For example, performance prediction can be interpreted from multiple components, offering a comprehensive understanding.
- Team performance optimization interpretation identifies influential network elements in the optimization results.
- Existing methods may identify influential edges and nodes in team member replacement, guiding favorable replacements.

Structuring unstructured data for knowledge mining

- Methodology for turning unstructured text-rich data into organized, structured knowledge.
- Promoting a distantly supervised, data-driven approach utilizing knowledge bases, dictionaries, and pre-trained language models.
- Components include information extraction, taxonomy construction, document classification, knowledge graph construction.

Structuring unstructured data for knowledge mining

- Taxonomy construction organizes concepts semantically, involving generating, refining, or expanding taxonomies.
- Weakly or distantly supervised methods for taxonomy generation and expansion, using set expansion, embedding-based clustering.
- Weakly supervised text classification with embedding methods, category-guided embedding, or pretrained language models.

Structuring unstructured data for knowledge mining

- Hierarchical multilabel text classification using taxonomies, generating category-distinctive keywords for supervision.
- Fine-grained information extraction via context-aware distant supervision, involving data-driven phrase mining and type inference.
- ClusType method optimizes type propagation with relation phrases and multiview relation phrase clustering.

Structuring unstructured data for knowledge mining

- Taxonomy-guided supervision and pretrained language models used for fine-grained entity recognition.
- Knowledge graph/information network construction involves distant or weak supervision, using taxonomy-guided text classification.
- Construction of local knowledge graphs from specific situations in massive text data to avoid confusion in merging different associations.

Data augmentation

- Challenges in data mining models due to limited labeled data in many domains, such as medical image processing.
- Introduction of a promising technique called data augmentation to address the issue of scarce training data.
- Data augmentation aims to enrich the training data set to improve the generalizability of data mining and machine learning models.

Data augmentation

- Two categories of data augmentation methods: basic methods and learning-based methods.
- Basic methods leverage the invariance property of data samples and include operations like flipping, rotation, color adjustment, and erasing part of images.
- Learning-based methods include augmentation policy learning, generative adversarial nets (GANs)-based methods, and adversarial training.

Data augmentation

- Augmentation policy learning involves automatically searching for the best augmentation policy through the definition of subpolicies.
- GANs-based methods use the adversarial training approach to handle transformations that basic augmentation methods cannot effectively cover.
- Adversarial training focuses on improving the robustness of models by introducing unnoticeable perturbations to the original data samples.

Data augmentation

- Comparison of methods: basic and policy learning methods require predefined operations, while GANs-based methods and adversarial training involve parameterized components learned from abundant data.
- Challenges in GANs-based methods include the difficulty of finding classes with plenty of data samples, especially in tasks like medical imaging processing.
- Subtle differences between adversarial training and data augmentation methods, such as the nature of perturbations and the goals of improving robustness and generalizability.

From correlation to causality

- Despite data mining successes, causality analysis lags behind correlation analysis.
- Correlation analyzes statistical associations, causality delves into cause-and-effect.
- Causality analysis challenges include discovery and inference tasks.
- Discovery involves understanding which variables to manipulate; inference quantifies resulting changes.

From correlation to causality

- Applications range from interpreting deep learning models to ensuring fairness.
- The taxonomy categorizes methods into discovery and inference.
- Inference methods assume observed confounders, including regression adjustment and propensity score.
- Handling unobserved confounders involves instrumental variables and front-door criterion.

From correlation to causality

- Advanced methods like neural networks and ensembles enhance causality inference.
- Current trends involve data-driven strategies, integrating deep learning, and tackling challenges.
- Open problems include dealing with complex treatments, exploring applications, and adapting to big data.

Network as a context

- Networks play a pivotal role in data mining and machine learning.
- The "Network-of-X" phenomenon links entities, datasets, or models via contextual networks.
- Examples include a network of time series, network of networks, and network of data mining models.
- In a network of time series, devices monitoring signals are connected, aiding precise abnormal activity detection.

Network as a context

- A network of networks involves domain-specific collaboration networks connected by a research domain network.
- Network of data mining models connects intercorrelated models, enhancing prediction performance.
- Future directions include improving contextual network construction, exploring diverse applications for X, and integrating with other data mining problems.

Auto-ML: methods and systems

- Data mining and machine learning are widely used but building models for diverse real-world tasks is time-consuming.
- Auto-ML addresses these challenges, classified into Automated Feature Engineering (Auto-FE) and Automated Model and Hyperparameter Tuning (Auto-MHT).
- Auto-FE includes RL-based methods constructing transformation graphs and EA-based approaches using trees to represent feature transformations.

Auto-ML: methods and systems

- In Auto-MHT, widely used techniques include BO, RL, and GA, focusing on hyperparameter tuning and model architectures.
- Key challenges for Auto-ML include the lack of benchmarks, efficiency concerns, incorporating human knowledge, interpretability, and exploring applications beyond classification and regression tasks.

Privacy-preserving data mining

- Privacy concerns arise in data mining, illustrated by a retail company accurately targeting a teen girl with pregnancy-related ads, causing privacy intrusion.
- Privacy leakage during data analytics poses risks to data owners, users, and service providers, leading to re-identification and anonymity breaches.
- Differential privacy is introduced as a safeguard, allowing aggregate derivations without compromising individual identification.

Privacy-preserving data mining

- K-anonymity is a method to generalize data attributes, but it is vulnerable to attacks like homogeneity attacks.
- Federated learning and federated analytics enable collaborative data mining without sharing sensitive information, providing a privacy-preserving framework.
- Societal and legal efforts, such as EU GDPR, HIPAA, and directives, contribute to privacy protection in data mining.

Human-algorithm interaction

- Human-algorithm interaction optimizes collaboration in data mining.
- Crowdsourcing uses human intelligence for tasks challenging for algorithms.
- Human-in-the-loop incorporates human intelligence in learning algorithms.

Human-algorithm interaction

- Machine-in-the-loop leverages learning algorithms to support humans.
- Human-machine teaming involves coordinated collaboration for superior performance.
- Challenges in human-machine teaming include modeling, predicting, and refining interactions.

Mining beyond maximizing accuracy

- Data mining facilitates automated decision making in various domains.
- Examples include social media recommending content and companies using data mining in recruiting.
- Beyond accuracy, fairness, interpretability, and robustness are crucial metrics in data mining.
- Algorithmic fairness tackles bias issues, categorized as group, individual, and counterfactual fairness.

Mining beyond maximizing accuracy

- Strategies involve preprocessing, optimization at training, and postprocessing.
- Interpretability is crucial for trust and safety; various methods, like summary statistics, are used.
- Robustness ensures consistent results; adversarial attacks and defense strategies are studied.
- Proactive methods make models robust before adversarial attacks; reactive methods detect after model construction.

Data mining for social good

- Data mining for social good applies techniques to address societal challenges.
- The lead water pipe detection project in Flint, Michigan, used machine learning to replace hazardous pipes.
- Various applications, from education to transportation, benefit from data mining models.

Data mining for social good

- Examples include identifying at-risk students, predicting epidemics, and modeling ride-sharing dynamics.
- Challenges include data scarcity, the need for domain-specific evaluation metrics, human-in-the-loop integration, and sustainable deployment.